Soup Experiment Tracker · Submission note

SHIP for the measured local profile. One app instance, one viewer per run, 50 concurrent runs at 10 points per run per second, and a 60-second workload. Both measured runs stored and delivered all 30,000 points without duplicates. Control-browser p99 was 1.61–1.68 s against a 2 s gate, leaving 0.32–0.39 s of headroom. These short runs leave long-run degradation unmeasured. Baseline runs

What the latency measures

The load driver stamps each batch with `perf_counter()` after the scheduled wait, just before building the JSON and starting the first POST. The same clock records the Playwright callback when a row first appears in the control Chromium DOM. The nearest-rank p99 uses 600 rows per minute. Scheduled batching is outside the timer. JSON serialization, HTTP, writing, SSE delivery and the browser observer are inside it. Load driver

The 1 s polling interval is a sleep after an empty read. It is one part of the trip from POST to DOM. On the one-viewer repeat, POST to acknowledgment p99 was 0.64 s and POST to HTTP SSE p99 was 1.20 s. These are separate distributions and cannot split the slowest DOM samples into stages. Baseline summary

DON’T SHIP at five viewers per run. The second profile delivered all 150,000 expected updates. DOM p99 rose to 2.65 s, driver POST p99 to 1.86 s and HTTP SSE p99 to 2.25 s. Sampled peak app CPU rose from 37.80% to 77.43% and database CPU from 22.02% to 33.55%. Write-pool wait p99 fell from 80.68 to 70.57 ms. Five-viewer summary

Measured extra work. In a separate 30 s diagnostic at each viewer count, PostgreSQL page reads rose from 3,044 to 14,607 (4.8×). Total SQL execution time rose from 0.44 s to 1.90 s while mean execution time per read stayed near 0.14 ms. This confirms repeated read work. Its share of the DOM tail remains unmeasured. SQL diagnostic

First change. Prototype one shared live reader per run while retaining per-viewer authorization and indexed replay from PostgreSQL. Repeat the fan-out workload and compare SQL reads and DOM p99. A wake-up signal alone would still leave one reader per subscription.

Consistency and isolation

The Chromium test combines duplicate frames, repeated batches, late points and a controlled disconnect. It compares the sent values, PostgreSQL and the final DOM. Another variant injects a `502` on reconnect. The recovery path reads persisted events after the last applied `seq`. Browser tests

Separate keys restrict reading and writing to the owning project. Tests cover the run list, snapshot, write, SSE and returning to the previous project’s URL through browser history. Foreign and unknown IDs have the same read and SSE error body. Protected responses use `Cache-Control: no-store`. API tests · Browser tests

Other leak paths. The database error handler returns generic text and logs the error class and SQLSTATE. Proxy or trace logs can expose headers or payloads if configured to record them. Identical error bodies still leave timing side channels possible. Isolation design

Silent failures and how they were checked

Full Compose test output

Failure	How it was checked
Events lost on reconnect	A controlled SSE disconnect holds the next request while the writer adds points. The final DOM matches input and PostgreSQL. Disabling replay leaves missing rows. Expected failure
Duplicates	A batch is retried with the same ID and then with a new ID. A browser variant receives a duplicate SSE frame. Two concurrent batches sharing a point both succeed. Removing the run lock produces a uniqueness error. Expected failure
Slow client and queue overflow	The stream reads up to 100 points at a time without a growing subscriber queue. The test stalls an ASGI data send and checks the 3 s timeout, release of the DB connection and replay after reconnect. Stream test
Partial writes	A test kills a separate writer process after metrics, cursor and receipt are written and before `COMMIT`. PostgreSQL shows an empty run and no receipt. Retrying the same batch inserts both points. Crash test
Old key after rotation	The old write key and read session are rejected after rotation. The idle stream rechecks after up to 1 s of sleep plus DB and scheduling time. With the production interval, the local test observed EOF 0.946 s after the rotation commit, within its 2 s gate. A fetched page or an authorized write may already be in flight. Rotation test

What the tests can miss. Multi-instance deployments, database failover, clock skew between hosts and other browsers are outside the setup. Controlled disconnect and stalled-send hooks can miss real TCP backpressure, keepalive stalls and bugs in the hooks themselves. Browser history traversal does not exercise an in-memory page restore or a misconfigured shared cache. The one-clock latency calculation avoids cross-host clock subtraction and does not model clock skew in production.

Unexpected result and remaining concern. An SSE shutdown flag carried over from one in-process test server to the next. Later streams closed as soon as they opened. Resetting the flag between fixtures fixed the test isolation. The five-viewer run still exceeded the latency gate even with every point delivered. The early failed load runs are preserved. Full report

AI use. Codex assisted with implementation, tests, load analysis and editing. The candidate identified the commit-order risk of a PostgreSQL sequence, the need for a deterministic race barrier and the failure mode after an SSE `502`. The resulting changes were checked with PostgreSQL and Chromium, including the two intentionally failing controls. AI-use log